MicroSeg pilot results

Cityscapes + Kaggle FPVCrosswalk (KPV) | 4 October 2026 | Bounded training run

HELD-OUT KPV DICE	HELD-OUT KPV IoU	TRAINING
0.000	0.000	5 epochs

Executive finding

The training pipeline completed, but this pilot did not produce usable Crosswalk predictions. Held-out real Crosswalk Dice, IoU, and pixel accuracy were zero in every epoch; held-out Crosswalk loss rose from 1.384 to 2.147. Do not simply extend this run.

Run configuration

Model	Four-class MicroSeg CNN; 96 × 96 RGB input
Training	5 epochs × 50 steps; batch size 8; seed 42
Runtime	WSL CPU; TensorFlow 2.21.0 (GPU initialization unavailable)
Cityscapes	2,975 train / 500 validation; 115 pairs had ignore-mask paths
KPV	242 real + 3,000 synthetic train; 58 held-out real validation
Sampling	35% Crosswalk batches; half of Crosswalk batches from real train
Validation	63 Cityscapes steps/epoch; held-out real KPV evaluation

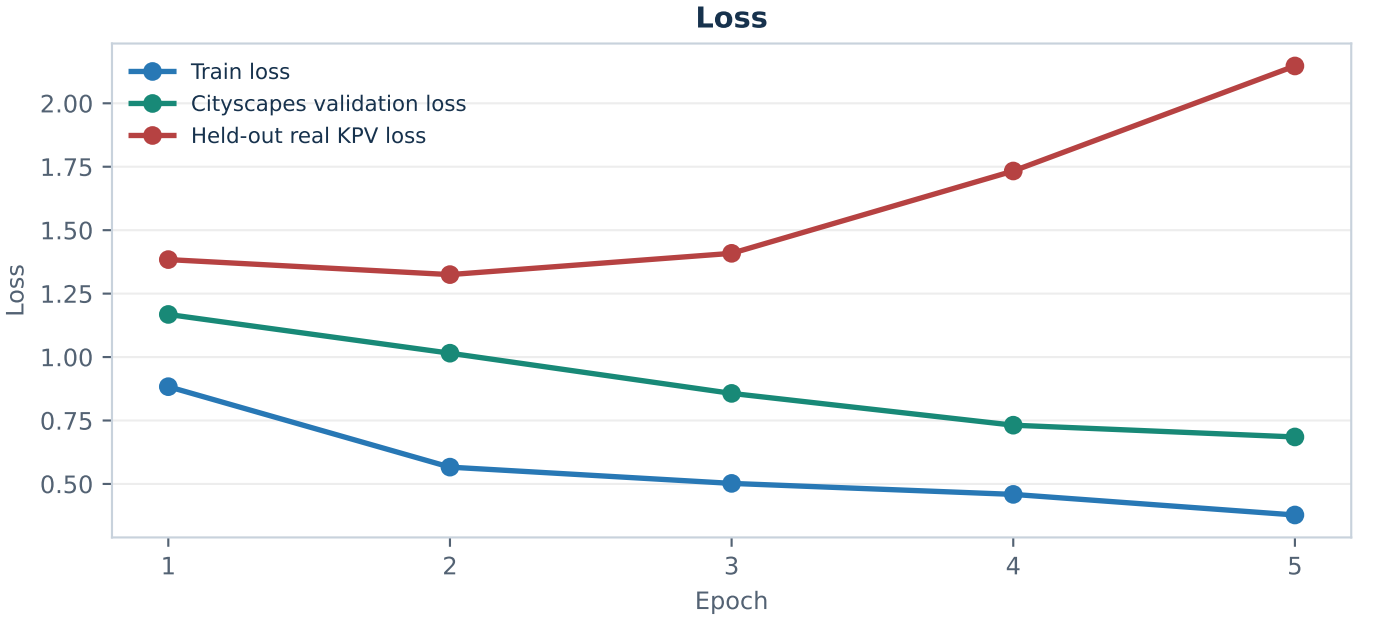
Interpretation

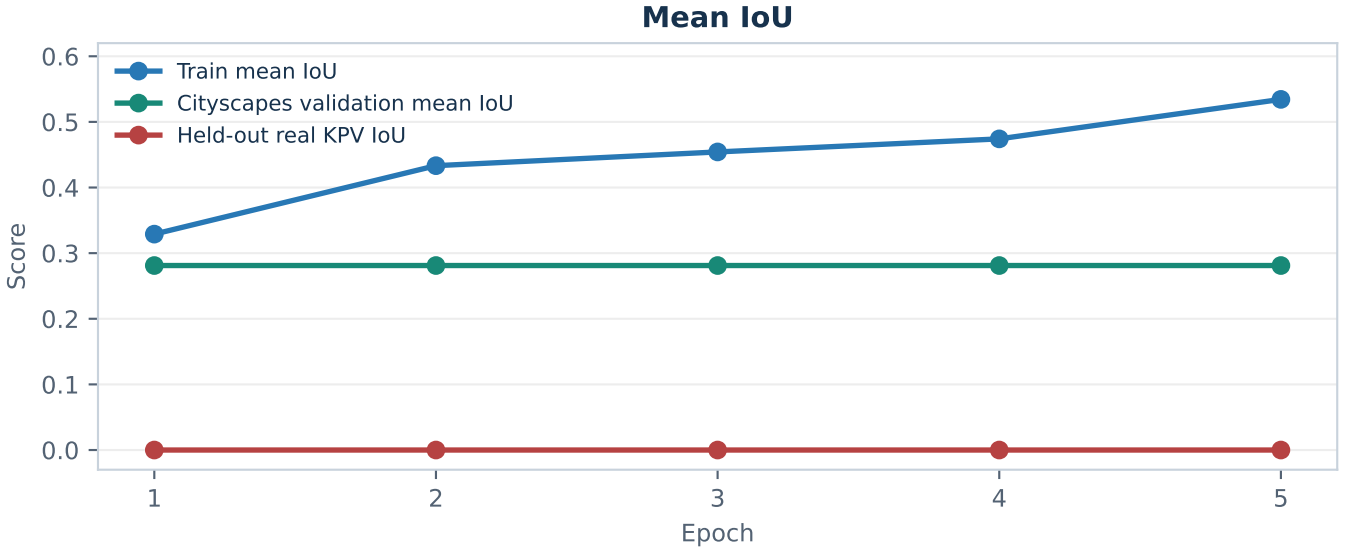
- Training loss fell from 0.883 to 0.378 and training mean IoU rose from 0.329 to 0.534, but held-out Crosswalk metrics stayed at zero. Training improved without evidence of Crosswalk generalization.
- Cityscapes validation mean IoU stayed at 0.281 and pixel accuracy at 0.843 across all epochs. The checkpoint-selection metric did not improve beyond epoch 1.
- KPV labels supervise positive Crosswalk pixels only. These held-out metrics are recall-oriented and do not measure false positives on unmarked pixels.
- Next: inspect predictions and class-wise behavior, then revise sampling/loss strategy before another run. This short pilot is not a final benchmark.

Source: persisted run configuration and history. Run: /home/nacsi/microseg_runs/pilot_cityscapes_kpv_20261004

Epoch-by-epoch metrics

Training curves and validation outcomes copied from the saved run history





Epoch	Train loss	Train mIoU	City val loss	City val mIoU	KPV val loss	KPV Dice / IoU
1	0.883	0.329	1.168	0.281	1.384	0.000 / 0.000
2	0.566	0.433	1.015	0.281	1.325	0.000 / 0.000
3	0.502	0.454	0.857	0.281	1.409	0.000 / 0.000
4	0.459	0.474	0.732	0.281	1.733	0.000 / 0.000
5	0.378	0.534	0.685	0.281	2.147	0.000 / 0.000

KPV validation uses positive-labeled pixels only; the metrics do not quantify false-positive performance.